

a 20 milestones span 1983–2025

- 1983 IP₃ releases Ca²⁺ from an intracellular store
- 1989 primary structure — it is the receptor
- 1989 purified receptor conducts Ca²⁺
- 1989 “similar to ryanodine receptor”
- 1990 P400 is the IP₃-binding protein
- 1991 Ca²⁺ dependence is bell-shaped
- 1991 subtypes differ by tissue and stage
- 1992 three receptors are found, not one
- 1995 release is built from quantal puffs
- 1996 *Itpr1* null mice are ataxic
- 2002 the IP₃-binding core is solved
- 2007 *ITPR1* deletion causes SCA15
- 2014 *ITPR2* loss abolishes sweating
- 2015 the tetramer is resolved by cryo-EM at 4.7 Å
- 2016 all four sites must be occupied
- 2018 paralogue structures are solved with Ca²⁺ and IP₃
- 2020 *ITPR3* variants cause CMT1J
- 2022 a ligand series maps activation and gating
- 2024 *ITPR3* variant causes multisystem disease
- 2025 one CMT1J variant recurs in nine families

● what it is ● how it works ● what goes wrong

b the 213 references this review cites are binned by year, with the milestones marked

